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afterGameEnds(RecursiveTicTacToeGame.RecursiveTicTacToeGameView) - Method in class edu.bu.labs.rttt.agents.RandomAgent

Agent - Class in edu.bu.labs.rttt.agents
A class representing an arbitrary agent in recursive tic tac toe.

Agent(PlayerType) - Constructor for class edu.bu.labs.rttt.agents.Agent
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applyFirstMove(Coordinate, Coordinate) - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame.RecursiveTicTacToeGameView
A method to apply the provided move (e.g. the `Coordinate` the current player wants to mark) on the "inner" game located at the other provided `Coordinate`.

applyMove(Coordinate) - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame

applyMove(Coordinate) - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame.RecursiveTicTacToeGameView
A method to apply the provided move (e.g. the `Coordinate` the current player wants to mark) on the current "inner" game.

applyMove(Coordinate, PlayerType) - Method in class edu.bu.labs.rttt.game.TicTacToeGame

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C

calculateOutcome() - Method in class edu.bu.labs.rttt.game.TicTacToeGame

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canAnyMoveBeMade() - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame.RecursiveTicTacToeGameView
This is an interesting edge-case of this game.

CELL_NUM_PIXELS - Static variable in class edu.bu.labs.rttt.game.Constants.Rendering.Cells.Dimensions

Cells() - Constructor for class edu.bu.labs.rttt.game.Constants.Rendering.Cells

CellType - Enum Class in edu.bu.labs.rttt.game
An enum representing the contents inside a square on a tic tac toe game.

changePlayer() - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame

clone() - Method in class edu.bu.labs.rttt.utils.Coordinate

Colors() - Constructor for class edu.bu.labs.rttt.game.Constants.Rendering.Cells.Colors

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Colors() - Constructor for class edu.bu.labs.rttt.game.Constants.Rendering.Player.Colors

Constants - Class in edu.bu.labs.rttt.game

Constants() - Constructor for class edu.bu.labs.rttt.game.Constants

Constants.Rendering - Class in edu.bu.labs.rttt.game
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Constants.Rendering.Cells - Class in edu.bu.labs.rttt.game

Constants.Rendering.Cells.Colors - Class in edu.bu.labs.rttt.game

Constants.Rendering.Cells.Dimensions - Class in edu.bu.labs.rttt.game

Constants.Rendering.Perimeter - Class in edu.bu.labs.rttt.game

Constants.Rendering.Perimeter.Colors - Class in edu.bu.labs.rttt.game

Constants.Rendering.Perimeter.Dimensions - Class in edu.bu.labs.rttt.game

Constants.Rendering.Player - Class in edu.bu.labs.rttt.game

Constants.Rendering.Player.Colors - Class in edu.bu.labs.rttt.game

Constants.Rendering.Player.Dimensions - Class in edu.bu.labs.rttt.game

Coordinate - Class in edu.bu.labs.rttt.utils

Coordinate(int, int) - Constructor for class edu.bu.labs.rttt.utils.Coordinate

D

didPlayerWin(PlayerType) - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame

didPlayerWin(PlayerType) - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame.RecursiveTicTacToeGameView
A method to determine if the specified `PlayerType` has won the recursive tic tac toe game.

didPlayerWin(PlayerType) - Method in class edu.bu.labs.rttt.game.TicTacToeGame

didPlayerWin(PlayerType) - Method in class edu.bu.labs.rttt.game.TicTacToeGame.TicTacToeGameView
A method to determine if the specified `PlayerType` has won this game.

Dimensions() - Constructor for class edu.bu.labs.rttt.game.Constants.Rendering.Cells.Dimensions

Dimensions() - Constructor for class edu.bu.labs.rttt.game.Constants.Rendering.Perimeter.Dimensions

Dimensions() - Constructor for class edu.bu.labs.rttt.game.Constants.Rendering.Player.Dimensions

draw(Graphics2D) - Method in class edu.bu.labs.rttt.rendering.GameRenderer

drawAllPerimeters(Graphics2D) - Method in class edu.bu.labs.rttt.rendering.GameRenderer

drawGame(Graphics2D) - Method in class edu.bu.labs.rttt.rendering.GameRenderer

drawInnerBoardPerimeter(Graphics2D, int, int, int, int) - Method in class edu.bu.labs.rttt.rendering.GameRenderer

drawInnerGame(Graphics2D, int, int, int, int, TicTacToeGame) - Method in class edu.bu.labs.rttt.rendering.GameRenderer

drawOuterBoardPerimeter(Graphics2D) - Method in class edu.bu.labs.rttt.rendering.GameRenderer

E

edu.bu.labs.rttt - package edu.bu.labs.rttt

edu.bu.labs.rttt.agents - package edu.bu.labs.rttt.agents

edu.bu.labs.rttt.game - package edu.bu.labs.rttt.game

edu.bu.labs.rttt.rendering - package edu.bu.labs.rttt.rendering

edu.bu.labs.rttt.traversal - package edu.bu.labs.rttt.traversal

edu.bu.labs.rttt.utils - package edu.bu.labs.rttt.utils

EMPTY - Enum constant in enum class edu.bu.labs.rttt.game.CellType

EMPTY_CELL_BACKGROUND_COLOR - Static variable in class edu.bu.labs.rttt.game.Constants.Rendering.Cells.Colors

equals(Object) - Method in class edu.bu.labs.rttt.utils.Coordinate

F

fromString(String) - Static method in class edu.bu.labs.rttt.utils.Coordinate

G

GamePanel - Class in edu.bu.labs.rttt.rendering

GamePanel(RecursiveTicTacToeGame) - Constructor for class edu.bu.labs.rttt.rendering.GamePanel

GameRenderer - Class in edu.bu.labs.rttt.rendering

GameRenderer(RecursiveTicTacToeGame) - Constructor for class edu.bu.labs.rttt.rendering.GameRenderer

getAgent(String, PlayerType) - Static method in class edu.bu.labs.rttt.Main

getAgentO() - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame

getAgentX() - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame

getAvailableFirstMoves() - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame.RecursiveTicTacToeGameView
A method to get all available moves that can be made on the first turn of the game.

getAvailableMoves() - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame.RecursiveTicTacToeGameView

A method to get all available `Coordinate`s that the current player could choose on the current "inner" game.

getAvailableMoves() - Method in class edu.bu.labs.rttt.game.TicTacToeGame.TicTacToeGameView
A method to get all available `Coordinate`s that the current player could choose on this game.

getCells() - Method in class edu.bu.labs.rttt.game.TicTacToeGame

getCellType(int, int) - Method in class edu.bu.labs.rttt.game.TicTacToeGame

getCellType(int, int) - Method in class edu.bu.labs.rttt.game.TicTacToeGame.TicTacToeGameView
A getter method to get the contents of a square specified by row/col indices.

getCellType(Coordinate) - Method in class edu.bu.labs.rttt.game.TicTacToeGame

getCellType(Coordinate) - Method in class edu.bu.labs.rttt.game.TicTacToeGame.TicTacToeGameView
A getter method to get the contents of a square specified by a `Coordinate`.

getChildren() - Method in class edu.bu.labs.rttt.traversal.Node
A method to get a list of Node objects representing all moves that could be made by the current player.

getCurrentGameCoord() - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame

getCurrentGameCoord() - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame.RecursiveTicTacToeGameView
A getter method to get the `Coordinate` of the "inner" game the current player must play on.

getCurrentPlayerType() - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame

getCurrentPlayerType() - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame.RecursiveTicTacToeGameView
A getter method to get the `PlayerType` of the player who'se turn it is.

getCurrentPlayerType() - Method in class edu.bu.labs.rttt.traversal.Node
A getter method to get the `PlayerType` of the current player (the one making a move this turn).

getDepth() - Method in class edu.bu.labs.rttt.traversal.Node
A getter method to get the depth of this node in the tree.

getFirst() - Method in class edu.bu.labs.rttt.utils.Pair

getGame() - Method in class edu.bu.labs.rttt.rendering.GameRenderer

getGame(int, int) - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame

getGame(Coordinate) - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame

getGameView(int, int) - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame.RecursiveTicTacToeGameView
A getter method to get the `TicTacToeGame.TicTacToeGameView` of an "inner" game specified by row/col indices.

getGameView(Coordinate) - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame.RecursiveTicTacToeGameView
A getter method to get the `TicTacToeGame.TicTacToeGameView` of an "inner" game specified by a `Coordinate`.

getIsGameOver() - Method in class edu.bu.labs.rttt.game.TicTacToeGame

getIsGameOver() - Method in class edu.bu.labs.rttt.game.TicTacToeGame.TicTacToeGameView
A getter method to get if this "inner" game has an outcome.

getLastMove() - Method in class edu.bu.labs.rttt.traversal.Node
A getter method to get the last move made (e.g. the move that produced the state stored in this Node).

getMaxNumMoves() - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame

getMaxNumMoves() - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame.RecursiveTicTacToeGameView
A getter method to get the maximum number of moves allowed in the game.

getMyPlayerType() - Method in class edu.bu.labs.rttt.agents.Agent
A getter method to get the `PlayerType` this Agent is controlling.

getMyPlayerType() - Method in class edu.bu.labs.rttt.traversal.Node
A getter method to get the `PlayerType` of the MAX player.

getNumCols() - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame

getNumCols() - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame.RecursiveTicTacToeGameView
A getter method to get the number of cols (x coordinate) for the recursive "outer" game.

getNumCols() - Method in class edu.bu.labs.rttt.game.TicTacToeGame

getNumCols() - Method in class edu.bu.labs.rttt.game.TicTacToeGame.TicTacToeGameView
A getter method to get the number of cols (x coordinate) for this "inner" game.

getNumMovesMade() - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame

getNumMovesMade() - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame.RecursiveTicTacToeGameView
A getter method to get the number of moves made in the game so far.

getNumRemainingCoords() - Method in class edu.bu.labs.rttt.game.TicTacToeGame

getNumRemainingCoords() - Method in class edu.bu.labs.rttt.game.TicTacToeGame.TicTacToeGameView
A getter method to get the number of empty squares remaining on this "inner" game.

getNumRows() - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame

getNumRows() - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame.RecursiveTicTacToeGameView
A getter method to get the number of rows (y coordinate) for the recursive "outer" game.

getNumRows() - Method in class edu.bu.labs.rttt.game.TicTacToeGame

getNumRows() - Method in class edu.bu.labs.rttt.game.TicTacToeGame.TicTacToeGameView
A getter method to get the number of rows (y coordinate) for this "inner" game.

getOppositePlayerType() - Method in class edu.bu.labs.rttt.traversal.Node
A method to get the `PlayerType` of the MIN player.

getOutcome() - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame

getOutcome() - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame.RecursiveTicTacToeGameView
A method to get the outcome of a game.

getOutcome() - Method in class edu.bu.labs.rttt.game.TicTacToeGame

getOutcome() - Method in class edu.bu.labs.rttt.game.TicTacToeGame.TicTacToeGameView
A getter method to get the outcome of this game.

getOutcome(int, int) - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame

getOutcome(int, int) - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame.RecursiveTicTacToeGameView
A getter method to get the outcome value of an "inner" game specified by row/col indices.

getOutcome(Coordinate) - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame

getOutcome(Coordinate) - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame.RecursiveTicTacToeGameView
A getter method to get the outcome value of an "inner" game specified by a `Coordinate`.

getRandom() - Method in class edu.bu.labs.rttt.agents.RandomAgent

getSecond() - Method in class edu.bu.labs.rttt.utils.Pair

getTerminalUtility() - Method in class edu.bu.labs.rttt.traversal.Node
A method to calculate the terminal utility of this Node.

getUtilityValue() - Method in class edu.bu.labs.rttt.traversal.Node
A getter method to get the utility value of this Node.

getView() - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame

getView() - Method in class edu.bu.labs.rttt.game.TicTacToeGame

getView() - Method in class edu.bu.labs.rttt.traversal.Node
A getter method to get the state of the game stored in this Node.

getXCoordinate() - Method in class edu.bu.labs.rttt.utils.Coordinate

getYCoordinate() - Method in class edu.bu.labs.rttt.utils.Coordinate

H

hashCode() - Method in class edu.bu.labs.rttt.utils.Coordinate

hashCode() - Method in class edu.bu.labs.rttt.utils.Pair

I

INNER_PERIM_NUM_PIXELS - Static variable in class edu.bu.labs.rttt.game.Constants.Rendering.Perimeter.Dimensions

isGameOver() - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame

isGameOver() - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame.RecursiveTicTacToeGameView
A method to determine if the game is over.

isInBounds(Coordinate) - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame

isInBounds(Coordinate) - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame.RecursiveTicTacToeGameView
A method to determine if the provided `Coordinate` is in-bounds (e.g. the location is valid).

isInBounds(Coordinate) - Method in class edu.bu.labs.rttt.game.TicTacToeGame

isInBounds(Coordinate) - Method in class edu.bu.labs.rttt.game.TicTacToeGame.TicTacToeGameView
A method to determine if the provided `Coordinate` is in-bounds (e.g. the location is valid).

isLegalMove(Coordinate) - Method in class edu.bu.labs.rttt.game.TicTacToeGame

isLegalMove(Coordinate) - Method in class edu.bu.labs.rttt.game.TicTacToeGame.TicTacToeGameView
A method to determine if the square specified by the provided `Coordinate` is available.

isLegalMove(Coordinate, Coordinate) - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame

isLegalMove(Coordinate, Coordinate) - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame.RecursiveTicTacToeGameView
A method to determine if the square specified by `moveCoord` is available within an "inner" game located at `gameCoord`.

isTerminal() - Method in class edu.bu.labs.rttt.traversal.Node
A getter method to get if this state is terminal or not (e.g. is the game over or not).

M

main(String[]) - Static method in class edu.bu.labs.rttt.Main

Main - Class in edu.bu.labs.rttt

Main() - Constructor for class edu.bu.labs.rttt.Main

makeFirstMove(RecursiveTicTacToeGame.RecursiveTicTacToeGameView) - Method in class edu.bu.labs.rttt.agents.Agent
An abstract method that child classes must override.

makeFirstMove(RecursiveTicTacToeGame.RecursiveTicTacToeGameView) - Method in class edu.bu.labs.rttt.agents.RandomAgent

makeFirstMove(RecursiveTicTacToeGame.RecursiveTicTacToeGameView) - Method in class edu.bu.labs.rttt.agents.SearchAgent
A method to calculate the first move of the game which contains two decisions: the @{link Coordinate} of the "inner" game to select and then the `Coordinate` of the square on that "inner" game to mark.

makeOtherMove(RecursiveTicTacToeGame.RecursiveTicTacToeGameView) - Method in class edu.bu.labs.rttt.agents.Agent
A n abstract method that child classes must override.

makeOtherMove(RecursiveTicTacToeGame.RecursiveTicTacToeGameView) - Method in class edu.bu.labs.rttt.agents.RandomAgent

makeOtherMove(RecursiveTicTacToeGame.RecursiveTicTacToeGameView) - Method in class edu.bu.labs.rttt.agents.SearchAgent
A method to calculate the move to make when this is not the first move in the game.

MARK_NUM_PIXELS - Static variable in class edu.bu.labs.rttt.game.Constants.Rendering.Player.Dimensions

markUnavailable(Coordinate) - Method in class edu.bu.labs.rttt.game.TicTacToeGame

N

Node - Class in edu.bu.labs.rttt.traversal
A class that represents a vertex in the game tree that adversarial search agents expand.

Node(RecursiveTicTacToeGame.RecursiveTicTacToeGameView, PlayerType, int) - Constructor for class edu.bu.labs.rttt.traversal.Node
A constructor to initialize a Node with the state of the game, who is the MAX player, and the depth of the node in the tree.

O

O - Enum constant in enum class edu.bu.labs.rttt.game.CellType

O - Enum constant in enum class edu.bu.labs.rttt.game.PlayerType

OUTER_PERIM_NUM_PIXELS - Static variable in class edu.bu.labs.rttt.game.Constants.Rendering.Perimeter.Dimensions

P

paintComponent(Graphics) - Method in class edu.bu.labs.rttt.rendering.GamePanel

Pair<K,V> - Class in edu.bu.labs.rttt.utils

Pair(K, V) - Constructor for class edu.bu.labs.rttt.utils.Pair

PERIM_COLOR_DEFAULT - Static variable in class edu.bu.labs.rttt.game.Constants.Rendering.Perimeter.Colors

PERIM_COLOR_O_WINNER - Static variable in class edu.bu.labs.rttt.game.Constants.Rendering.Perimeter.Colors

PERIM_COLOR_TIE - Static variable in class edu.bu.labs.rttt.game.Constants.Rendering.Perimeter.Colors

PERIM_COLOR_X_WINNER - Static variable in class edu.bu.labs.rttt.game.Constants.Rendering.Perimeter.Colors

Perimeter() - Constructor for class edu.bu.labs.rttt.game.Constants.Rendering.Perimeter

Player() - Constructor for class edu.bu.labs.rttt.game.Constants.Rendering.Player

PLAYER_O_COLOR - Static variable in class edu.bu.labs.rttt.game.Constants.Rendering.Player.Colors

PLAYER_X_COLOR - Static variable in class edu.bu.labs.rttt.game.Constants.Rendering.Player.Colors

PlayerType - Enum Class in edu.bu.labs.rttt.game

playTurn() - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame

POPULATED_CELL_BACKGROUND_COLOR - Static variable in class edu.bu.labs.rttt.game.Constants.Rendering.Cells.Colors

R

RandomAgent - Class in edu.bu.labs.rttt.agents

RandomAgent(PlayerType) - Constructor for class edu.bu.labs.rttt.agents.RandomAgent

RecursiveTicTacToeGame - Class in edu.bu.labs.rttt.game

RecursiveTicTacToeGame() - Constructor for class edu.bu.labs.rttt.game.RecursiveTicTacToeGame

RecursiveTicTacToeGame(RecursiveTicTacToeGame.RecursiveTicTacToeGameView) - Constructor for class edu.bu.labs.rttt.game.RecursiveTicTacToeGame

RecursiveTicTacToeGame.RecursiveTicTacToeGameView - Class in edu.bu.labs.rttt.game
A class that represents a read-only immutable view into the state of the recursive tic tac toe game.

Rendering() - Constructor for class edu.bu.labs.rttt.game.Constants.Rendering

S

search(Node) - Method in class edu.bu.labs.rttt.agents.SearchAgent
An abstract method that child classes must override.

SearchAgent - Class in edu.bu.labs.rttt.agents
A class representing an arbitrary agent in recursive tic tac toe.

SearchAgent(PlayerType) - Constructor for class edu.bu.labs.rttt.agents.SearchAgent
A constructor to initialize a SearchAgent instance.

setCurrentGameCoord(Coordinate) - Method in class edu.bu.labs.rttt.game.RecursiveTicTacToeGame

setUtilityValue(double) - Method in class edu.bu.labs.rttt.traversal.Node
A setter method to set the utility of this Node.

T

TicTacToeGame - Class in edu.bu.labs.rttt.game

TicTacToeGame() - Constructor for class edu.bu.labs.rttt.game.TicTacToeGame

TicTacToeGame(int) - Constructor for class edu.bu.labs.rttt.game.TicTacToeGame

TicTacToeGame(TicTacToeGame.TicTacToeGameView) - Constructor for class edu.bu.labs.rttt.game.TicTacToeGame

TicTacToeGame.TicTacToeGameView - Class in edu.bu.labs.rttt.game
A class that represents a read-only immutable data structure containing the state of an individual tic tac toe game (also referred to as an "inner" game by `RecursiveTicTacToeGame`).

toString() - Method in class edu.bu.labs.rttt.utils.Coordinate

TOTAL_HEIGHT - Static variable in class edu.bu.labs.rttt.rendering.GameRenderer

TOTAL_WIDTH - Static variable in class edu.bu.labs.rttt.rendering.GameRenderer

U

UNAVAILABLE - Enum constant in enum class edu.bu.labs.rttt.game.CellType

UNAVAILABLE_CELL_BACKGROUND_COLOR - Static variable in class edu.bu.labs.rttt.game.Constants.Rendering.Cells.Colors

V

valueOf(String) - Static method in enum class edu.bu.labs.rttt.game.CellType
Returns the enum constant of this class with the specified name.

valueOf(String) - Static method in enum class edu.bu.labs.rttt.game.PlayerType
Returns the enum constant of this class with the specified name.

values() - Static method in enum class edu.bu.labs.rttt.game.CellType
Returns an array containing the constants of this enum class, in the order they are declared.

values() - Static method in enum class edu.bu.labs.rttt.game.PlayerType
Returns an array containing the constants of this enum class, in the order they are declared.

X

X - Enum constant in enum class edu.bu.labs.rttt.game.CellType

X - Enum constant in enum class edu.bu.labs.rttt.game.PlayerType

A C D E F G H I M N O P R S T U V X
All Classes and Interfaces | All Packages | Constant Field Values | Serialized Form

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